

Second Semester

Name of Paper: Personality and Soft Skills Development

Course No.: BCA/2/CC/07

Marks Scale: **100** marks (End Sem. Exam: 75+Internal: 25)

Credit: **4**
(3- 1- 0)

Unit 1: Introduction to Personality Development

(10L)

Basics of Personality Development and its importance: Definition, Components and Scope, Communication Skills and Personality Development.

Unit 2: Grooming Personality

(10L)

Motivation, Leadership skills and team building, Goal setting, Time Management and Effective planning.

Unit 3: Elements of a Letter

(10L)

Email Communication: introduction, techniques for writing effective e-mail, email etiquette
Letter Writing: Job Application letters, writing Resume.

Unit 4: Business Letter

(10L)

Business Letters: Letter of Enquiry, quotations, order and acknowledgement letters, complaint and adjustment letters.

Unit 5: Effective Writing

(10L)

Writing Abstracts and Summaries: Report Writing- Structure and Layout, Elements of Structure, Front Matter, Main Body, Back Matter; Laboratory Reports.

Recommended Books:

1. Rajiv K Mishra: *Personality Development*, Rupa& Co.
2. Wallace and Masters: *Personal Development for Life Work*, 8th Edition, Thomson
3. Goodwill Publishing House: *All About Body Language*
4. R.C Sharma-Krishna Mohan: *Business Correspondence and Report Writing*, Tata McGraw Hill Pub. Co. Ltd.
5. Aruna Koneru: *Professional Communication*, Tata McGraw Hill Pub. Co. Ltd.

Second Semester
Name of Paper: Discrete Mathematics
Course No.: BCA/2/CC/08

Marks Scale: **100** marks (End Sem. Exam: 75+Int.: 25)

Credit: **5**
(4- 1- 0)

Unit-1:

(10L)

Set Theory: Sets; Different type of sets and problems, Roster form (or Tabulation Method), Venn Diagrams; Boolean Algebra, General Properties of Boolean Algebra, Boolean functions, Boolean Switching.

Unit-2:

(10L)

Logic and Proposition Calculus: Statements and Notation, Connectives, Negation, Conjunction, Disjunction, Statement Formulas and Truth Tables, Conditional and Biconditional Statements; Tautologies and Contradictions; Logical Equivalence; Logical Implication, Predicates.

Unit-3:

(10L)

Permutations and Combinations: Factorial Notation, Fundamental Principles of counting, Permutations (including practical problems); Combinations (including practical problems)

Binomial Theorem: Binomial Theorem for Positive Index; Observations in a Binomial Expansion; the beginning, coefficient, independent term.

Unit-4:

(10L)

Number Theory: Divisibility, Division algorithm, Greatest Common Divisor, Existence and uniqueness of GCD, Relatively Prime Integers, LCM, Euclid's Lemma, Unique factorization, Congruence of Integers, Euler's phi-function, Fermat's theorem, Euler's theorem.

Unit-5:

(10L)

Graph Theory: Graph, Simple graph, multigraph, Directed graph, Weighted graph; Bipartite graphs; Eulerian and Hamiltonian graphs; Subgraphs.

Tree: Spanning Tree; minimum spanning tree; planar and non-planar graphs; Graph colouring; Adjacency Matrix and Incidence Matrix, Shortest Path Algorithm – Prims and Kruskals.

Recommended Books:

1. MK Gupta: *Discrete Mathematics*, Krishna Prakashan Media(P) Ltd. Meerut
2. J.P. Tremblay and R. Manohar: *Discrete Mathematical Structures with Applications to Computer Science*; Tata McGraw-Hill Publishing Co. Ltd., New Delhi.
3. Narsingh Deo: *Graph Theory with Applications to Engineering and Computer Science*, Prentice Hall of India Private Limited.
4. A.R. Vasishtha and A.K. Vasishtha: *Modern Algebra*, Krishna Prakashan Media(P) Ltd. Meerut

Second Semester

Name of Paper: Programming Language through C
Course No.: BCA/2/CC/09

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **4**
(3- 1- 0)

Unit-1:

(10L)

Overview of C: Importance of C, sample C programs, basic structure of C programs, Programming style, executing C program. Character set, C tokens, keywords and identifiers, constants, variables, data types, declaration of variables, assigning value to variable, defining symbolic constants.

Operators and Expression: Operators - arithmetic, relational, logical, assignment, increment-decrement, conditional, bit-wise and special. Arithmetic expressions, evaluation of expressions, precedence of arithmetic operators, type conversions in expressions, operator precedence and associativity.

Managing Input and Output Operators: Formatted input/output-more about printf() and scanf() functions. Unformatted input – getchar(), getch(), getche(), gets(). Unformatted output – putchar(), puts().

Unit-2:

(10L)

Overview of C: Decision making with IF statement: simple IF statement, the IF-ELSE statement, nesting of IF...ELSE statements, the ELSE IF ladder. The switch statement, break, continue, goto statements, enum.

Decision Making and Looping: Looping statements - WHILE, DO.... WHILE and FOR. Nesting and Jumps in loops, Infinite loop

Unit-3:

(10L)

Functions: Definition, prototype of a function, standard library functions, User -Defined Functions, Need for user-defined functions, return values and their types, category of functions: no arguments and no return values, arguments but no return values, arguments with return values. Handling of non-integer functions, calling a function-call by value and call by reference, recursion and iteration, storage classes of variables automatic, static, register and external, their scopes and lifetime.

Unit-4:

(10L)

Arrays: Definition, declaration and initialization: One-dimensional arrays, two-dimensional arrays. Multidimensional arrays.

Pointers: Definition, accessing the address of variable, declaring and initializing pointers, accessing a variable through its pointer, pointers and arrays, arrays of pointers, pointers and functions.

Unit-5:

(10L)

Structures and Unions: Structure definition, structure initialization, user-defined data types, arrays of structures, arrays within structures, structures within structures, structures and functions, pointers and structures. Self-referential structures, union, typedef.

File Management in C: File concept, various operations on files – Defining, opening, closing, input/output operations on files, command line arguments.

Recommended Books:

1. E. Balagurusamy : *Programming in ANSI C*, Tata McGraw Hill Publications (2004).
2. Ashok. N.Kamthane : *Programming with ANSI and Turbo C*, Pearson Education (2006).
3. B. Kernigan and D. Ritchie: *The ANSI C Programming Language*, PHI Publications (1988).
4. Byron Gottfried: *Programming with C*, Tata McGraw Hill Publications (2010).

Second Semester
Name of Paper: Accounting and Financial Management
Course No.: BCA/2/CC/10

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **4**
(3- 1- 0)

Accounting:

Unit-1: (10L)

Introduction to Accountancy: definition, objectives, advantages & limitations; Accounting Principles, Concepts & Conventions, Double entry System; Rules of Debit & Credit – Modern (American) Approach & Traditional rules (British Approach), Journal; Accounting equation.

Ledger Accounts: Meaning, classifications, ledger posting from journal entries, ledger posting from Cash book, Accounting for Cash - Cash book (three column Cash book).

Unit-2: (10L)

Trial Balance: Meaning, special features and objectives; Preparation of trial balance.

Financial statement(with adjustment): Meaning of Trading, Profit & Loss a/c and Balance Sheet, *Adjustments:* Closing stock, outstanding expenses, prepaid expenses, Accrued income, advance income, bad debts, provision for bad and doubtful debt, provision for discount on debtors and creditors, Depreciation, interest on capital, interest on drawing, interest on loans.

Unit-3: (10 L)

Ratio Analysis: Meaning of Accounting ratios, objectives and limitations. Types of ratios and their usefulness – Liquidity Ratio, Profitability Ratio, Efficiency ratio, solvency ratios.

Financial Management:

Unit-4: (10 L)

Introduction to Management Accounting: Meaning, objectives, nature & scope, advantages & limitations of Management accounting. Differences between Financial Accounting and Management Accounting, Management Accounting and Cost Accounting. Management Accountant's position, roles and responsibilities.

Standard costing: Meaning, objective, advantages & limitations of Standard Costing.

Unit- 5: (10 L)

Budgeting: Definition, Budget Vs Forecasts, and Essentials of budgeting. Types of Budget – Functional, Master, Fixed, flexible Budget and zero-based budget. (Theory and simple problems), **Budgetary Control:** Meaning, objectives, advantages and limitations.

Unit Costing: Preparation of Cost Sheet and Tender Price Statement.

Recommended Books:

1. N. Maheshwari: *Cost and Management Accounting*, Sultan Chand & Sons (2013).
2. M. N. Arora : *Cost & Management Accounting*, Vikas Publishing House Pvt Limited (2009).
3. Basu & Das : *Practice in Accountancy*, Vol-I, Rabindra Library (2004).
4. N. Maheswari : *Advance Accounting*, Vikas Publication House Pvt. Ltd (2009).
5. R. L. Gupta : *Advance Accounting*, Sultan Chand (1999).

Second Semester
Name of Paper: Programming in C Lab
Course No.: BCA/2/CC/11

Marks Scale: **100** marks (End Sem. Exam: 75+Internal: 25)

Credit: **3**
(0- 0- 3)

A. Record Book	(10 marks)
B. Viva Voce	(15 marks)
C. Practical	(50 marks)

List of Minimum Practical to be performed:

Unit-1: C Programming Basics

- 1) Marks obtained by a student in five different subjects are input through the keyboard. Calculate the percentage and display on the screen. Assume that the maximum marks in each subject is 100.
- 2) Temperature of your city in Fahrenheit degrees is input through the keyboard. Write a program to convert this temperature into Centigrade degrees.

Unit-2: Operators, Expressions, Branching and Looping

- 3) Write a C program for demonstration of nested if...else statement to find out the greatest of three numbers input by the user.
- 4) Percentage of your attendance is input through the keyboard. Write C program to calculate and display internal mark based on percentage of attendance as given below:
 - if attendance ≥ 95 , internal mark is 5
 - if attendance ≥ 90 && < 95 , internal mark is 4
 - if attendance ≥ 85 && < 90 , internal mark is 3
 - if attendance ≥ 80 && < 85 , internal mark is 2
 - if attendance ≥ 75 && < 80 , internal mark is 1
 - if attendance $< 75\%$, internal mark 0 and not eligible to appear in examination.
- 5) Write a C program which input three numbers and display the largest number using conditional operator.
- 6) Write a C program using switch to display day of the week by entering the corresponding number. For example, if input is 1, it will display Sunday and so on.
- 7) Write a C program to print the Fibonacci series up to n terms using while loop.
- 8) Write a C program to check whether the given number is prime or not.

Unit-3: Arrays and String Handling Functions

- 9) Write a program to perform Linear Search of any value in an array and display the position of the value if found.
- 10) Write a program using C to perform binary search for any item in an ordered array and display the position of the item if found.
- 11) Elements of two matrices are input by the user. Write a C program to find and display the addition and subtraction of the two matrices.
- 12) Using C programming language, write a program to find the transpose of a matrix.
- 13) Write C program to find the multiplication of two matrices using two dimensional array.
- 14) Write a program in C to accept lines of text and then display the number of alphabets, digits and special characters.

Unit-4: Functions and Structures

- 15) Write a recursive function program using C to find the factorial value of any number entered through the keyboard.
- 16) Write a C program to demonstrate functions call by value and call by references.
- 17) Write a C program to reverse the accepted number using function.
- 18) Write a C program to convert decimal number to its equivalent binary number using function.
- 19) Define a structure called student with data members RollNo, Name and Percentage. Also write two functions getdata() to input data and printdata() to display the data.

Unit-5: Pointers and File Management in C

- 20) Write a C program to read and print student details such as roll number as integer, name as character and percentage as float using structure pointer.
- 21) Write a C program to create file "Student.txt" which will save student information such as student roll number, name and total marks for n students and display lists of students on screen by reading from the same file.
- 22) Write a program in C to accept lines of text to save into a file and then read from the same file to display the number of alphabets, digits and special characters on the screen.

Note: *The above lists of experiments are suggestive of the standard. Department may modify, add or delete any experiment whenever it considered necessary*

Second Semester
Name of Paper: Tally ERP 9.0 Lab
Course No.: BCA/2/CC/12

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **3**
(0- 0- 3)

A. Record Book	(10 marks)
B. Viva Voce	(15 marks)
C. Practical	(50 marks)

PART A:

1. Meaning of Tally, importance, benefits of tally.
2. Creating
 - a. *New company*
 - b. *Security Controls*
 - c. *Groups*
 - d. *Ledger*
 - e. *Voucher Type*
3. Modifying
 - a. *New company*
 - b. *Security Controls*
 - c. *Groups*
 - d. *Ledger*
 - e. *Voucher Type*
4. Voucher Entry
5. Generating Profit & Loss Account, Trial Balance and Balance Sheet
6. Backup & Restore.

PART B:

1. GST in tally ERP.9.0
2. Application of GST in company
3. How to import GST.
 - a) *Ledger creation*
 - b) *Multi GST sales transaction*
 - c) *Tax invoice*
 - d) *Receipt entry for advance payment.*

Note: *The above lists of experiments are suggestive of the standard. Department may modify, add or delete any experiment whenever it considered necessary*